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02-23R/2024 (PHY2104C)

2024

PHYSICS

(Core)

Paper : PHY2104C

(Wave and Optics)

Full Marks : 45

Time : 1½ hours

**The figures in the margin indicate
full marks for the questions.**

SECTION-I

1. Answer the following questions :

(a) A particle is subjected to two mutually perpendicular simple harmonic motion

$X = 2 \sin \omega t$ and $Y = 3 \sin \left(\omega t + \frac{\pi}{4} \right)$. The path of the particle will be 1

- (i) ellipse
- (ii) straight line
- (iii) parabolic
- (iv) circular

Contd.

(b) Why longitudinal waves are also called pressure waves? 2

(c) What is stationary wave? Show that the distance between two successive

nodes is $\frac{\lambda}{2}$, where λ is wavelength of the wave. 2+6=8

OR

A particle is subjected simultaneously to two simple harmonic motions of same periods but of different amplitudes and phases in perpendicular direction. Find the expression for the resultant motions. Draw Lissajous figures if phase difference between two vibrations are

$\pi, \frac{\pi}{2}, \frac{\pi}{4}$. 5+3=8

SECTION-II

2. Answer **either** (a) **or** (b) :

(a) (i) Explain Huygen's principle. 2

(ii) What do you mean by constructive and destructive interference? Obtain the conditions for constructive and destructive interference. 2+5=7

(iii) In an interference experiment, the fringe width is obtained as 0.2 mm. If the whole experiment is done under water ($\mu = 4/3$) without changing the setup, what will be the new fringe width? 3

(b) (i) What are fringes of equal thickness and fringes of equal inclination? 2

(ii) Obtain the conditions for dark and bright fringes in case of interference produced in a parallel thin film for reflected series. Hence explain the colour of a thin film. 5+3=8

(iii) The central fringe of Newton's rings is dark. Why? 2

SECTION-III

3. Answer **either** (a) **or** (b) :

(a) (i) Write the differences between Fresnel-type and Fraunhofer-type diffractions. 3

- (ii) What are Fresnel's half period zones? Why are they so called? Show that area of all half period zones drawn on a plane wavefront with respect to a point are same.

$$1+2+5=8$$

- (b) (i) Discuss the intensity distribution on the screen due to Fraunhofer diffraction due to a single slit. 8

- (ii) Obtain an expression for the width of the central maximum in case of Fraunhofer diffraction due to a single slit. 3

SECTION-IV

4. (a) Which of the following phenomena of light proves that light is a transverse wave? 1

(i) Interference of light

(ii) Diffraction of light

(iii) Polarization of light

(iv) Photoelectric effect

- (b) State Malu's law. 2

- (c) State Brewster's law. Show that the reflected rays and refracted rays are perpendicular to each other when the unpolarised light are incident at the angle of polarization. $1+4=5$

OR

What is quarter wave plate ? Calculate the thickness of quarter wave plate and half wave plate. $1+4=5$

- ✓(d) What is double refraction of light ? Explain it with Huygen's principle.

$1+2=3$